Will Buziak

willbuziak@gmail.com — will_buziak@mines.edu

Background

Computer & Mechanical Engineer - Computer Science PhD student

Education

Colorado School of Mines
Doctor of Philosophy
Field of Study: Computer Science
Expected Graduation: Spring 2028

University of Tennessee, Knoxville
Bachelor of Science
Major: Mechanical Engineering
Minor: Computer Science
Graduated: May 2024

Research Interests

Compuational Physics, Quantum Computation & Hardware, Microarchitecture Design, Multi-Physics Modeling

Experience

Bahar Lab
Graduate Research Assistant — Aug 2024 - Present

- Hybrid Quantum-Classical Algorithm development
- Secure Memory Computer Architecture Design for RISC-V Trusted Execution Environments
- Simulation Verification and Modification using the gem5 simulator

References: Dr. Iris Bahar (Mines), Dr. Tamara Silbergleit Lehman (UCB), Dr. Pooja Rao (NVIDIA)

Los Alamos National Lab
Graduate Student Intern — June 2026 - August 2026

- Security research on LLMs and hardware resource pattern analysis

References: Dr. Michael Teti (LANL)

Neuromorphic Computing TENNLab
Undergraduate Research Assistant — Dec 2022 - Aug 2024

- Mathematical modeling on a joint project with ORNL Research Scientists utilizing a neuromorphic controller on a dual fuel diesel-ammonia combustion engine

- Robotic design for swarming robotic applications emphasizing onboard low-power electronics
- Event-based camera data processing for object recognition utilizing speed filtering and clustering algorithms within the TENNLab framework

References: Dr. James Plank (UTK), Dr. Catherine Schuman (UTK), Dr. Charles Rizzo (UTK), Dr. Bryan Maldonado (ORNL), Dr. Brian Kaul (ORNL)

Electrochemical Energy Conversion and Storage Lab
Undergraduate Research Assistant — Aug 2022 - May 2024

- Multi-physics modeling for hydrogen electrolyzer research applications
- Designed a web-based interactive user interface for the visualization of Electric Vertical Take-Off and Landing (eVTOL) vehicle energy storage and power delivery system requirements

References: Dr. Matthew Mench (UTK), Dr. Douglas Aaron (UTK), Dr. Anirban Roy (NREL)

Eck-Lectric Industries
Mechanical Engineering Intern — Oct 2022 - Dec 2022

- Assisted in original product design for patent development

References: Fred Martin, Paul Eck

Shaw Industries
Process Engineering Co-op — May 2022 – Aug 2022

- Led process improvement projects in a manufacturing environment with a focus on waste optimization, automation and safety

References: Ashley Muench

IEEE / ACM Invited Talks

- YARCH '25 Workshop hosted by ASPLOS-EUROSYS 2025
- IEEE Denver Computer, Information Theory, and Robotics (CIR) - Spring 2025

Awards / Scholarships

- C-MAPP Scholar 2025-2026
- EnergyTech University Prize 2023 Bonus Prize finalist

Certificates

- IBM Qiskit Global Summer School 2025 - Quantum Excellence

Computer Skills

Languages: C/C++, Rust, Python, Java, HTML/CSS, MATLAB, RISC-V
Design Software: VHDL, Verilog / SystemVerilog, Solidworks, Onshape
Operating Systems: Linux/Unix, Windows Suite
Version Control: git (Github / Bitbucket / Gitlab)
Single Board Computers: Raspberry Pi 4/5, Raspberry Pi Pico, Arduino
Architecture Simulators: Gem5, QEMU
Quantum SDKs: Qiskit, D-Wave, CUDA-Q
Software: RTL Design, Mathematical & Multi-physics Modeling, Simulators and Emulation Frameworks
Quantum Computing: Quantum Programming, Architecture & Theoretical Quantum Computation

Additional Skills

Fabrication: Woodworking, Metal-Working, Additive Manufacturing, Soldering
Textiles: Crochet

Achievements:

Ultra-Running: Tejas Trails River's Edge - 50K (May 2026)

Contact

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- Github: github.com/wbuziak
- Personal Website: https://wbuziak.github.io/wbuziak/